

BEN HARDY

Nashville, TN | (517) 817-6322 | benjamin.mi.hardy@gmail.com | [Portfolio](#) | [LinkedIn](#) | [Google Scholar](#)

PROFILE

Application engineer working where RF and electromagnetic simulation meets agentic AI. At DesignForge, builds and evaluates an agentic copilot that drives Ansys HFSS/AEDT: the benchmark and evaluation harness that track accuracy across releases, multi-agent orchestration, a versioned knowledge corpus, and MMIC agent skills that turn hours of manual design work into minutes. Before that, nearly three years as a customer-facing application engineer at Remcom, an EM simulation software company: weekly demos for RF engineers at Apple, Meta, Amazon, Google and Rohde & Schwarz, customized tutorials, an industry webinar, deals at about 2× the company's average size, surrogate-model antenna optimization, and Fusion, an AI-native application with paying customers. Ph.D. in physics (Vanderbilt), including RF hardware for 15.2 T MRI.

WORK EXPERIENCE

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|---------------------|---|
| Mar 2026 – Present | <p>Senior Application Engineer, DesignForge
<i>Agentic AI copilot for RF/microwave design</i></p> <ul style="list-style-type: none">• Benchmark design. Designed the company's atomic-prompt benchmark (HFSS scenarios graded against gold AEDT reference projects) and a passives library of gold AEDT designs. Showed the domain agent was faster and far more token-efficient than a frontier coding agent on the same model.• Evaluation harness. Built and ran the harness that drives Ansys AEDT end to end to test how an agentic system uses the tool: a fresh project per run, Langfuse traces of every tool call, and token and cost tracking. Tracked accuracy across releases.• Ablation studies. Removed harness context to measure its value; removing the knowledge layer substantially lowered pass rate.• Multi-agent orchestration. Orchestrated multiple agents on parallel AEDT instances, including agentic "races" toward fixed design goals and unattended overnight runs.• Agentic memory. Built a versioned knowledge corpus that sessions promote verified lessons into, gated by lint and per-version tool-fact probes.• MMIC skills. Built MMIC agent skills for RF designers: spiral inductor L/Q extraction to MDIF, interdigital capacitor tuning to a target C and Q, and GDS-to-HFSS import with stackup, turning hours of manual work into minutes. Chained the skills to build a working three-pole GaAs MMIC bandpass filter.• Quality assurance. Led QA: code review of MCP server development, skill testing, protocols for skill hardening and skill QA, and hands-on manual QA. Supplied the measured benchmark and skill-timing results for the company's capability materials.• Outreach. Ran the customer database through an agentic system for outreach; created LinkedIn content. |
| Jun 2023 – Mar 2026 | <p>Solution/Application Engineer, Remcom Inc., Nashville, TN (remote)</p> <ul style="list-style-type: none">• Customer education. Weekly demos to RF engineers at Apple, Meta, Amazon, Google, Rohde & Schwarz and others; customized tutorials; LinkedIn content with 200k+ total impressions (example post).• Sales results. Deals closed at ~2× the company average size; 20% of deals touched drove 36% of revenue (strategic accounts); covered 42% of enterprise deals over \$50K.• Product feedback. Documented 75+ feature requests and bug reports.• Fusion. Built an automation tool that integrates all of Remcom's tools into one platform: an AI-native application built with Cursor, with paying customers.• Surrogate modeling. Led a project to optimize antenna geometry with surrogate models, working toward multi-objective 3D antenna geometry optimization. |

- **Neural network for breathing anomalies.** Built a TensorFlow network that detects breathing waveforms from radar phase, with an automated data pipeline (Python, batch, JavaScript); reached <5% MSE. Invited talk at ISAP 2025 [1].
- **GNSS satellite simulator.** Links a satellite's position relative to any latitude/longitude to a Wireless InSite project to compute carrier-to-noise over time for devices under test.

Aug 2017 –
Jun 2023

Graduate Research Assistant, Vanderbilt University, Nashville, TN
Imaging and hardware development for MRI

- Built a cryogenic chamber housing a Tx/Rx surface loop for imaging, improving SNR by a factor of 2 [3].
- Customized pulse sequence programming (C) on a Bruker 15.2 T MRI scanner; measured the effects of diffusion on signal and point spread function at resolution beyond 8 μm in phantoms and cell samples [4].
- Authored 3 first-author publications; coauthored 4 publications; authored 9 conference proceedings.

SKILLS

Agentic systems: Agentic memory systems, evaluation harness development, multi-agent orchestration, MCP, benchmark design, ablation studies, LLM observability (Langfuse).

AI coding agents: Claude Code, Cursor, Grok, OpenAI Codex, Gemini.

Languages: Python, MATLAB, C++, C, Bash, JavaScript, SLURM, Git, Qt, L^AT_EX.

Simulation: Ansys AEDT/HFSS, PyAEDT automation, CST, XFDTD, WaveFarer, Wireless InSite.

Hardware: Keysight network analyzers, SolidWorks, Eagle, KiCad, Formlabs.

RF & microwave: Waveguides, array optimization, receiver chains, on-body antennas.

EDUCATION

Jun 2023

Ph.D. Physics, Vanderbilt University, Nashville, TN
Focus: Magnetic Resonance Imaging | Advisor: Dr. John Gore
Thesis: [Advancing Ultra High Field Magnetic Resonance Microscopy](#)

May 2017

B.Sc. Physics and Mathematics, Bowling Green State University, Bowling Green, OH
Dean's Honor List | CGPA 3.87/4.00

TALKS

2025

Webinar: "Advancing Radar Sensing Simulation to Include Human Detection & Micro-Doppler in Complex Environments," International Wireless Industry Consortium, Jun 18, 2025. [Watch on YouTube](#)

2025

Invited talk, [International Symposium on Antennas and Propagation \(ISAP 2025\)](#), Fukuoka.

SELECTED PUBLICATIONS

- [1] **B.M. Hardy**, S. Mukherjee and T. Chawla, "Synthetic Radar Return-based Neural Network for Detecting Breathing Anomaly," 2025 *Int. Symp. Antennas Propag. (ISAP)*, Fukuoka, Japan, 2025, pp. 1–2. doi.org/10.23919/ISAP63122.2025.11362043
- [2] S. Mukherjee, **B.M. Hardy**, G.J. Skidmore, et al., "Animating Vital Signs in Radar Simulations: Comparing Physical Optics Against 28.5 GHz Channel Measurements," *Proc. IEEE RadarConf*, 2024. doi.org/10.1109/RadarConf2458775.2024.10548317
- [3] **B.M. Hardy**, G. Drake, S. Chai, et al., "A Cryogenic Tune and Match Circuit for Magnetic Resonance Microscopy at 15.2 T," *J. Magn. Reson. Open*, 2024. doi.org/10.1016/j.jmro.2024.100147
- [4] **B.M. Hardy**, Y. Zhu, K.D. Harkins, et al., "Experimental Demonstration of Diffusion Limitations on Resolution and SNR in MR Microscopy," *J. Magn. Reson.*, Jul. 2023. doi.org/10.1016/j.jmr.2023.107479